

Birla Institute of Technology and Science-Pilani, Hyderabad Campus
Second Semester 2023-2024
Comprehensive Exam

Part-A

Course No CS F364 Course Title: Design and Analysis of Algorithms
Type: Closed Total Time: 1 hr Total Marks: 30 Date: 12/05/24

Name: _____ ID: _____

Instructions: There are 3 pages of Part-A question paper. Attempt all the questions. Do not overwrite. Please submit Part-A and take Part-B question paper. Q9 and 15 are fill in the blanks question and all others are MCQ.

1. (2 points) Assume you have two positive functions f and g such that $f(n)$ is in $O(g(n))$. For each of the following statements, decide which one is TRUE:
- A. $2^{f(n)}$ is $O(2^{g(n)})$. B. $f(n)^2$ is $O(g(n)^2)$. C. $f(n) = O(f(n)^2)$. D. $g(n) = \Omega(g(n)^2)$.

1. _____

2. (2 points) Let $T(n)$ be the recurrence relation defined as follows:

$$\begin{aligned} T(0) &= 1, \\ T(1) &= 2, \text{ and} \\ T(n) &= 5T(n-1) - 6T(n-2) \text{ for } n \geq 2 \end{aligned}$$

Which one of the following statements is TRUE?

- A. $T(n) = \Theta(2^n)$. B. $T(n) = \Theta(n2^n)$. C. $T(n) = \Theta(3^n)$. D. $T(n) = \Theta(n3^n)$.

2. _____

3. (2 points) Which one of these is not a divide and conquer algorithm for sorting:

- A. Heapsort. B. Mergesort. C. Quicksort. D. None of the above

3. _____

4. (2 points) Given an unsorted array of n distinct elements, you want to find the set of $\log n$ elements: those at positions $1, 2, 4, 8, 16, \dots, n/2$ if the array is sorted. In other words, find the smallest element, second smallest element, fourth-smallest element and so on, terminating with the median element. Consider that we have an algorithm to find k^{th} smallest in an array of size n using $\Theta(n)$ time. Assuming n is a power of 2, how fast can you find all these $\log n$ elements?

- A. $\Theta(\log n)$. B. $\Theta(n)$. C. $\Theta(n \log n)$. D. $\Theta(n^2)$.

4. _____

5. (2 points) Consider the following C functions:

```
1 int foo(int n)
2 {
3     if (n < 6)
4         return 1;
5     else
6         return foo(n-1) + foo(n-3) + foo(n-5);
7 }
```

Suppose we modify the above function `foo()` and store the values of `foo(i)`, $0 \leq i < n$ as and when they are computed. With this modification, the time complexity of `foo()` is significantly reduced. What is the time complexity of modified `foo()`?

- A. $O(n!)$. B. $O(n^2)$. C. $O(n)$. D. $O(1)$.

5. _____

6. (2 points) Suppose you need to code up a problem solved via Dynamic Programming in a language that does not support recursion. Which one of the following approaches is the most efficient that you can implement?

- A. Top down with memoization B. Top down C. Bottom up D. Brute force

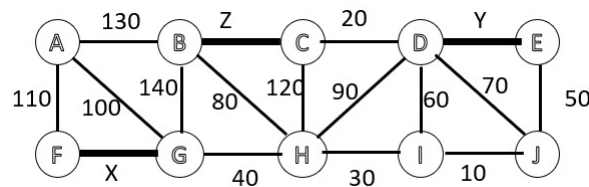
6. _____

7. (2 points) Which of the following statements is NOT TRUE?

- A. Greedy algorithms are easy to design.
 B. Proving correctness of a greedy algorithm is usually not straightforward.
 C. An algorithm which make greedy choices is always correct.
 D. Greedy algorithms' running time is straightforward to analyze.

7. _____

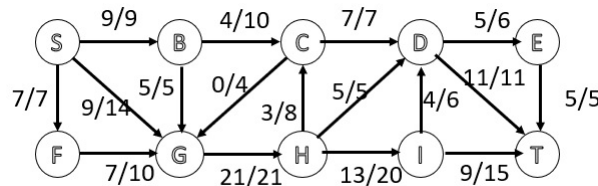
8. (2 points) Suppose that a MST of the following edge-weighted graph contains the edges with weights X , Y , and Z . What is the least upper bound of X , Y , and Z ?



- A. $X \leq 110, Y \leq 70, Z \leq 40$ B. $X \leq 110, Y \leq 60, Z \leq 80$
 C. $X \leq 110, Y \leq 60, Z \leq 120$ D. $X \leq 140, Y \leq 60, Z \leq 80$

8. _____

9. (2 points) Consider the following flow network and feasible flow f from from the source vertex S to the sink vertex T . What is the value of the maximum flow? _____



10. (2 points) Which one of the following is true in a $s - t$ flow network with multiple minimum $s - t$ cuts?

- A. Any maximum flow saturates exactly one minimum $s-t$ cut
 B. Any maximum flow saturates all minimum $s-t$ cuts
 C. There are maximum flows that saturate no minimum $s-t$ cut.
 D. Any maximum flow saturates all but one minimum $s-t$ cuts

10. _____

11. (2 points) Assuming $P \neq NP$, which of the following is true?

- A. $NP\text{-Complete} = NP$ B. $NP\text{-Complete} \cap P = \phi$ C. $NP\text{-Hard} = NP$ D. $P = NP\text{-Complete}$

11. _____

12. (2 points) Which of the following statements are true?

1. The problem of determining whether there exists a cycle in an undirected graph is in NP.
2. If a problem A is in NP, there exists a polynomial time algorithm to compute a solution to the problem A .
3. Given a pair of vertices u and v in a simple undirected graph $G = (V, E)$, longest simple path between u and v can be computed in $O(n^4)$ time where n is the number of vertices in the graph.

A. 1 and 2. B. 1 and 3. C. 2 and 3. D. 1 only.

12. _____

13. (2 points) Which one of the following is true?

- A. We have not proved the existence of a problem in NP that is also in P.
- B. We have not proved the existence of a problem in P that is also in NP.
- C. We have not proved the existence of a problem that is NP-hard that is also NP-complete.
- D. We have not proved the existence of a problem that is NP-complete that is also in P.

13. _____

14. (2 points) Which of the following options match the given statement

Statement: The algorithm that use the random number as an auxiliary input to reduce the expected running time or memory usage, but always terminate with a correct result in a bounded amount of time.

A. Las Vegas Algorithm B. Monte Carlo Algorithm C. Atlantic City Algorithm D. None of the above

14. _____

15. (2 points) Consider the process of making a graph randomly. We start with n nodes and join each pair by an undirected edge, independently with probability p . If $p = \frac{c}{n-1}$ for some constant $c \in \mathbb{N}, 0 < c < n$, what is the expected degree of a node? _____

END OF PAPER

TOTAL MARKS OBTAINED:

RECHECK REQUEST: